SHTC equations with Material Metric

Seminar — Selected Problems in Mathematical Modelling

Michael Monček

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Charles University · Mathematical Institute

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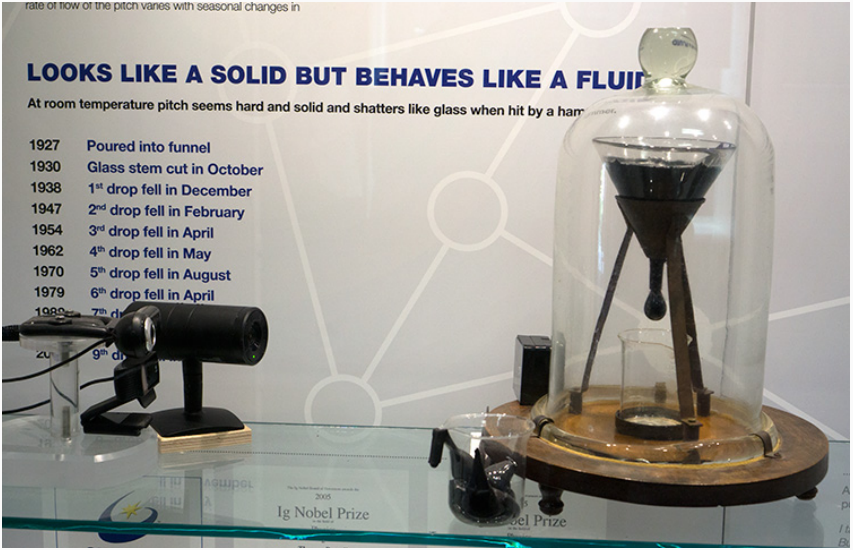
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Question: Can we construct a continuum theory that can handle the fluid-solid spectrum continuously — without switching frameworks at an arbitrary threshold?



Source: https://www.theandytchannel.com/wp-content/uploads/2019/08/pitch7-1.jpg

SHTC

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- $\tau \to 0+$ ideal fluids
- $0 < \tau < +\infty$ viscous fluids
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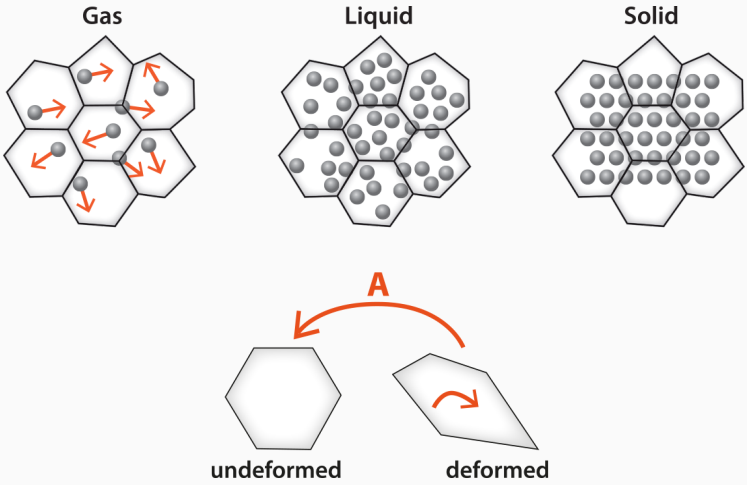
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The fluidity is introduced via relaxation of distortion field **A**.

SHTC



SHTC

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{1}$$

Momentum equation:

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\mathbf{v} \otimes \rho \mathbf{v} - \mathbf{S}) = 0 \tag{2}$$

Equation for distortion:

$$\frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{A} + \mathbf{A} \nabla \mathbf{v} = -\frac{1}{\theta} E_{\mathbf{A}} \tag{3}$$

Equation for energy:

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot (\mathbf{v} \rho E - \mathbf{S}\mathbf{v}) = 0 \tag{4}$$

$\mathbf{A}$ is the distortion field, $\mathbf{S}$ is the stress tensor, $\theta = \theta(\rho, \tau, \mathbf{A}) \geq 0$ is a *relaxation function*, $E = E(\rho, \mathbf{v}, \mathbf{A}) = \epsilon(\rho, \mathbf{A}) + \frac{1}{2}\|\mathbf{v}\|^2$ is total energy density.

Roadmap

Hutter & Pavelka Kincl et al. Bachelor's Thesis Particle distortion

Material Metric Formulation (Master's thesis)

Motivation and **previous work**

Hutter & Pavelka

A satisfies the following identity:

$$\mathbf{A}\mathrm{d}\mathbf{x} - \mathrm{d}\mathbf{X} = \mathbf{0}$$

we can use it as *an alternative definiton*.

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$$\mathbf{A}(\mathbf{x}) = \mathrm{argmin}_{\mathbf{A}} \sum_{a,b} w(\mathbf{x} - \mathbf{x}_a) w(\mathbf{x} - \mathbf{x}_b) \|\mathbf{A}\mathbf{x}_{ab} - \mathbf{X}_{ab}\|^2$$

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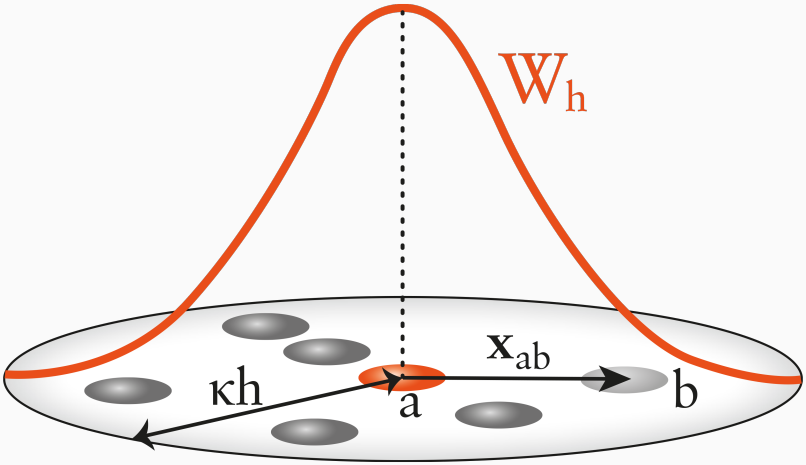
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This is a conceptual shift from evolving distortion in time to *computing it from particle data*.


W_h
x_{ab}
κh
a
b

Roadmap

Hutter & Pavelka Kincl et al. Bachelor's Thesis Particle distortion

Material Metric Formulation (Master's thesis)

Particle definition of distortion field **A**

Kincl et al.

A full unified SPH-SHTC formulation for both fluids and solids in a single framework, implemented in *Julia* as SmoothedParticles.jl.

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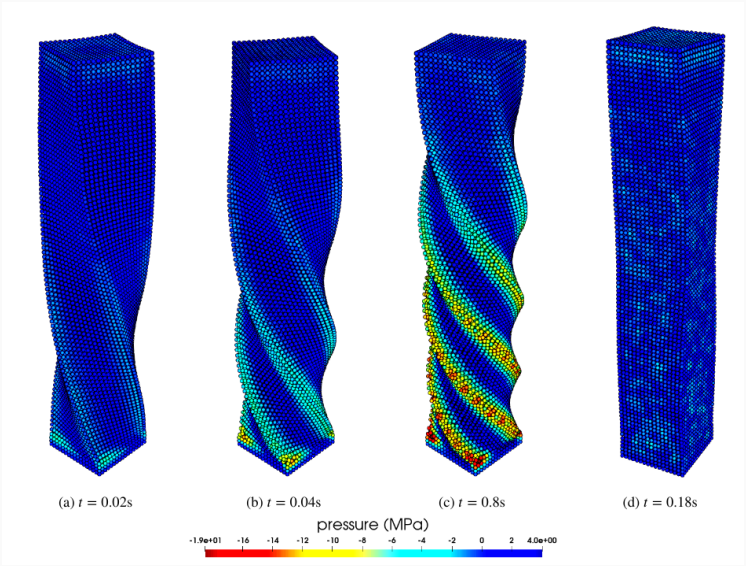
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However we do evolve $\mathbf{A}$ in time, by first approximating the velocity gradient $\mathbf{L}$ and then integrating:

$$\frac{\mathrm{d}\mathbf{A}}{\mathrm{dt}} = -\mathbf{A}\mathbf{L} - \frac{1}{\theta}\mathbf{E}_{\mathbf{A}}$$



Source: *Ondřej Kincl, Ilya Peshkov, Michal Pavelka, Václav Klika, Unified description of fluids and solids in Smoothed Particle Hydrodynamics, Applied Mathematics and Computation, Volume 439, 2023, 127579, ISSN 0096-3003, https://doi.org/10.1016/j.amc.2022.127579.*

Roadmap

Hutter & Pavelka **Kincl et al.** Bachelor's Thesis Particle distortion

Material Metric Formulation (Master's thesis)

Unified SPH-SHTC method for fluids and solids

Distortion field evolved in time

Bachelor's Thesis

An attempt to simulate fluid-structure interaction using the SHTC-SPH model.

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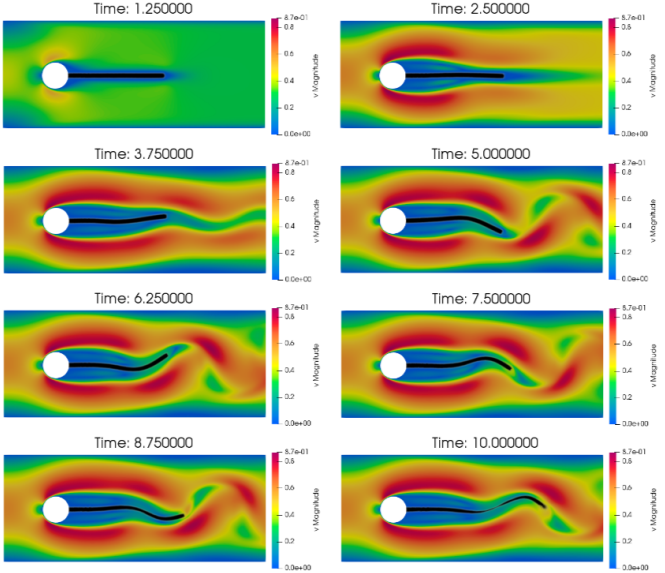
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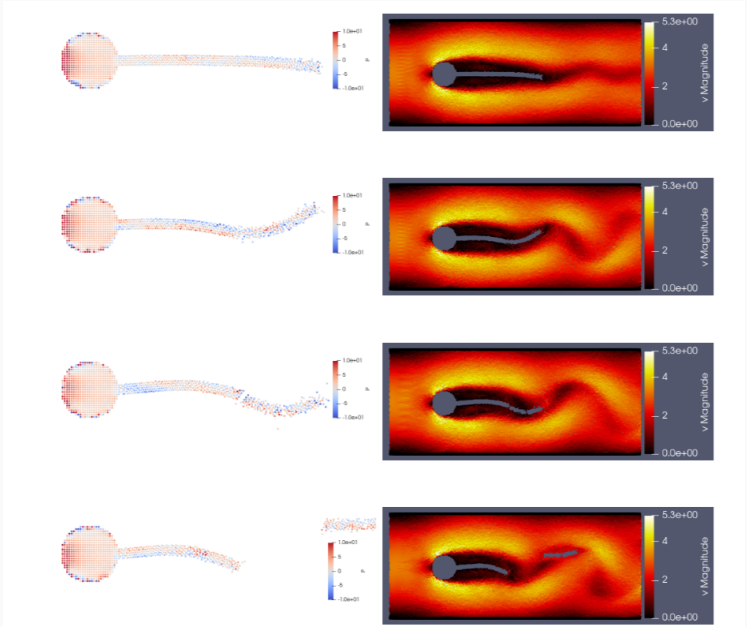
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Open research question: What causes the failure?



Source: *FARA, Jakub. Fluid-structure interaction between blood and dissipating artery wall. Diplomová práce, vedoucí Tůma, Karel. Praha: Univerzita Karlova, Matematicko-fyzikální fakulta, Matematický ústav UK, 2020.*



Source: *MONČEK, Michael. Modelling Fluid-Structure Interaction using Smoothed Particle Hydrodynamics. Bakalářská práce, vedoucí Pavelka, Michal. Praha: Univerzita Karlova, Matematicko-fyzikální fakulta, Matematický ústav UK, 2025.*

Roadmap

Hutter & Pavelka Kincl et al. **Bachelor's Thesis** Particle distortion

Material Metric Formulation (Master's thesis)

FSI simulation attempt. Turek-Hron flag tears.
Failure at the fluid-solid interface — not a numerical artifact.

Particle distortion in SPH

Implementation of the Hutter-Pavelka formulation — links it to the Kincl framework.

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Preliminary result: one order higher convergence.

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Preliminary result: one order higher convergence.

Question: How to encode plastic deformation into this definition?

Roadmap

Hutter & Pavelka Kincl et al. Bachelor's Thesis **Particle distortion**

Material Metric Formulation (Master's thesis)

Numerical implementation of Hutter-Pavelka
One order higher convergence than in Kincl et al.

Material Metric Formulation

Plasticity traditionally:
$\mathbf{F} = \mathbf{F}_e \mathbf{F}_p$ and $\mathbf{A} = \mathbf{F}_e^{-1}$

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Question: What if plasticity is not a decomposition of deformation, but a change in the material itself?

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Proposition: Treat the material body as a Riemannian manifold $(\mathcal{M}, \mathbf{g})$.

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Plasticity = irreversible evolution of $\mathbf{g}$.

Material Metric Formulation

This yields a clean conceptual separation from classical SHTC model.
In **Metric SHTC** we have:

Variable	Meaning
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g	Material internal state

Dissipation in **g** opens *new physics* unavailable in the classical formulation: plastic evolution of natural configuration, damage and microcracking, phase transformation, biological remodelling, structural relaxation in glasses,...

Two channels of dissipation

Dissipation in distortion

Classical SHTC:

$$\partial_t \mathbf{A} = -\mathbf{A}\mathbf{L} - \frac{1}{\theta} E_{\mathbf{A}}$$

Distortion loses memory of the reference mapping. Describes viscous or plastic flow.

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Dissipation in material metric

Metric SHTC:

$$\partial_t \mathbf{g} = -v \cdot \nabla \mathbf{g} - \frac{1}{\theta} E_{\mathbf{g}}$$

The natural microstructure evolves irreversibly. Describes internal state changes.

Roadmap

Hutter & Pavelka Kincl et al. Bachelor's Thesis Particle distortion

Material Metric Formulation (Master's thesis)

Plastic deformation encoded in an evolving material metric.
Distortion and metric relaxed independently.

Goals

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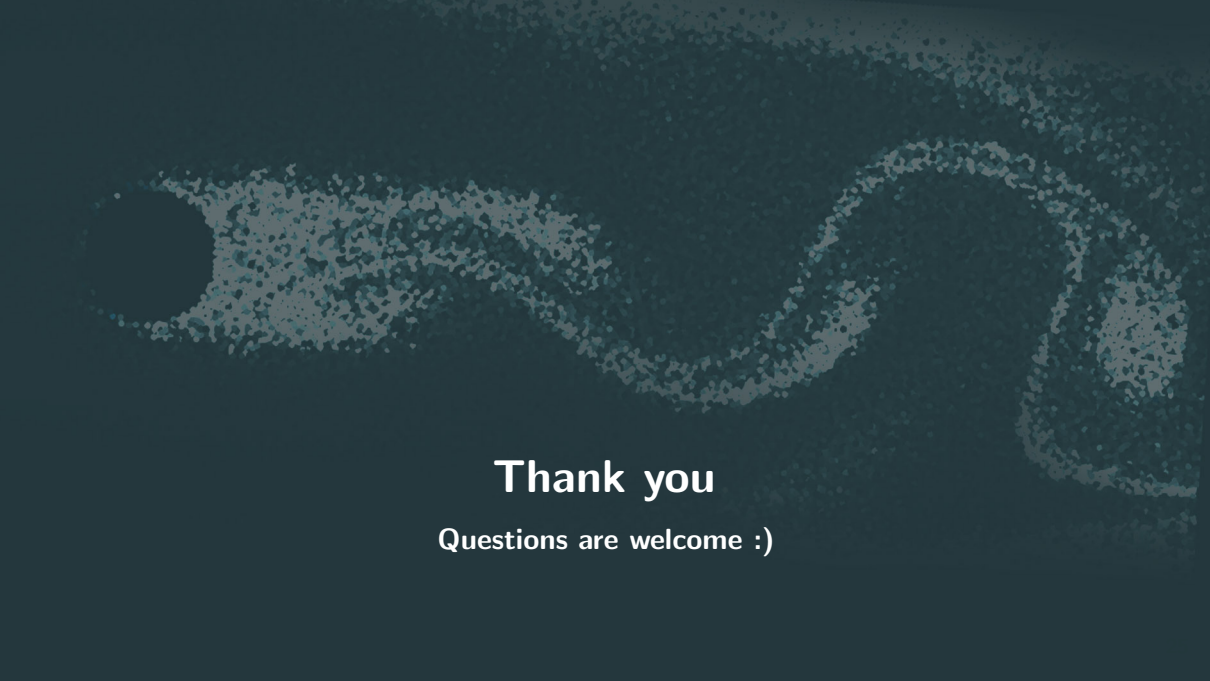
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3. Investigate properties and applications of the new formulation.
Anisotropy encoded in the metric, complex material behaviour, and biological applications as a longer-term horizon.


Thank you
Questions are welcome :)